

Abstract

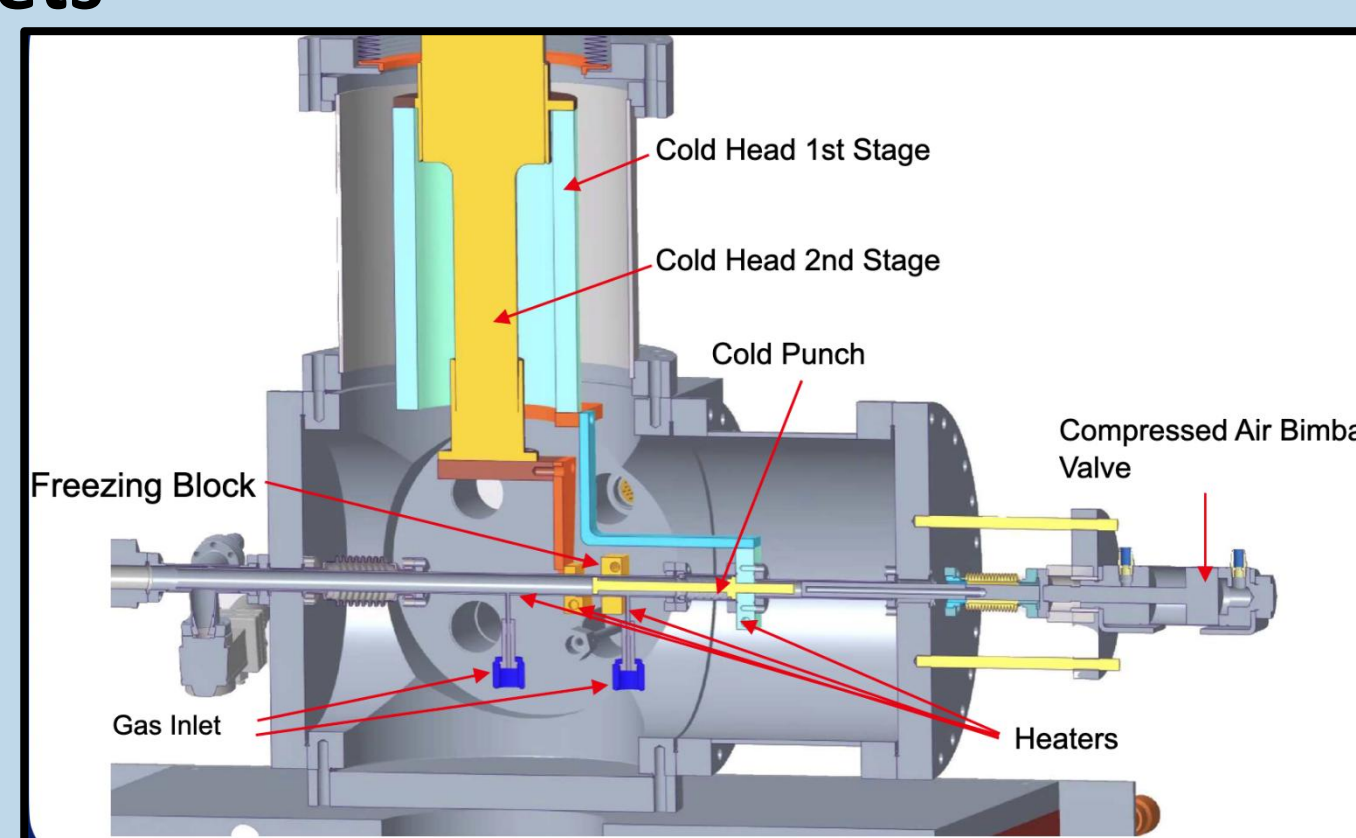
Among all the techniques used to mitigate plasma disruptions, Pellet Injection (PI) has become one of the most reliable and versatile methods. The Pellets at Columbia (PAC) test stand works towards understanding and applying the ablation of cryogenic pellets used for PI. PAC utilizes various diagnostic and operational systems to create and ablate these pellets, such as the Microwave Cavities (MCs) and Neon Gas System. This poster discusses these systems in detail and how they advance understand of Pellet Injection.

Motivation

- PAC test stand is intended to collect direct ablation data under high energy particle flux
- PI can be used as a fueling, disruption mitigation, or plasma control technique
- Direct pellet ablation has not been empirically studied in the fusion field, making PAC a unique and necessary test stand

Operation

- The Injection chamber and ablation chamber are rough-vacuum while making and shooting pellets
- The Hydrogen pellets are made and sheared from inner barrel with the force of a mechanical punch
- Pellets accelerate through the barrel into the ablation chamber to be ablated by the e-beam
- Diagnostic systems (camera, MCs) are then used to determine various traits of the pellets



Hardware

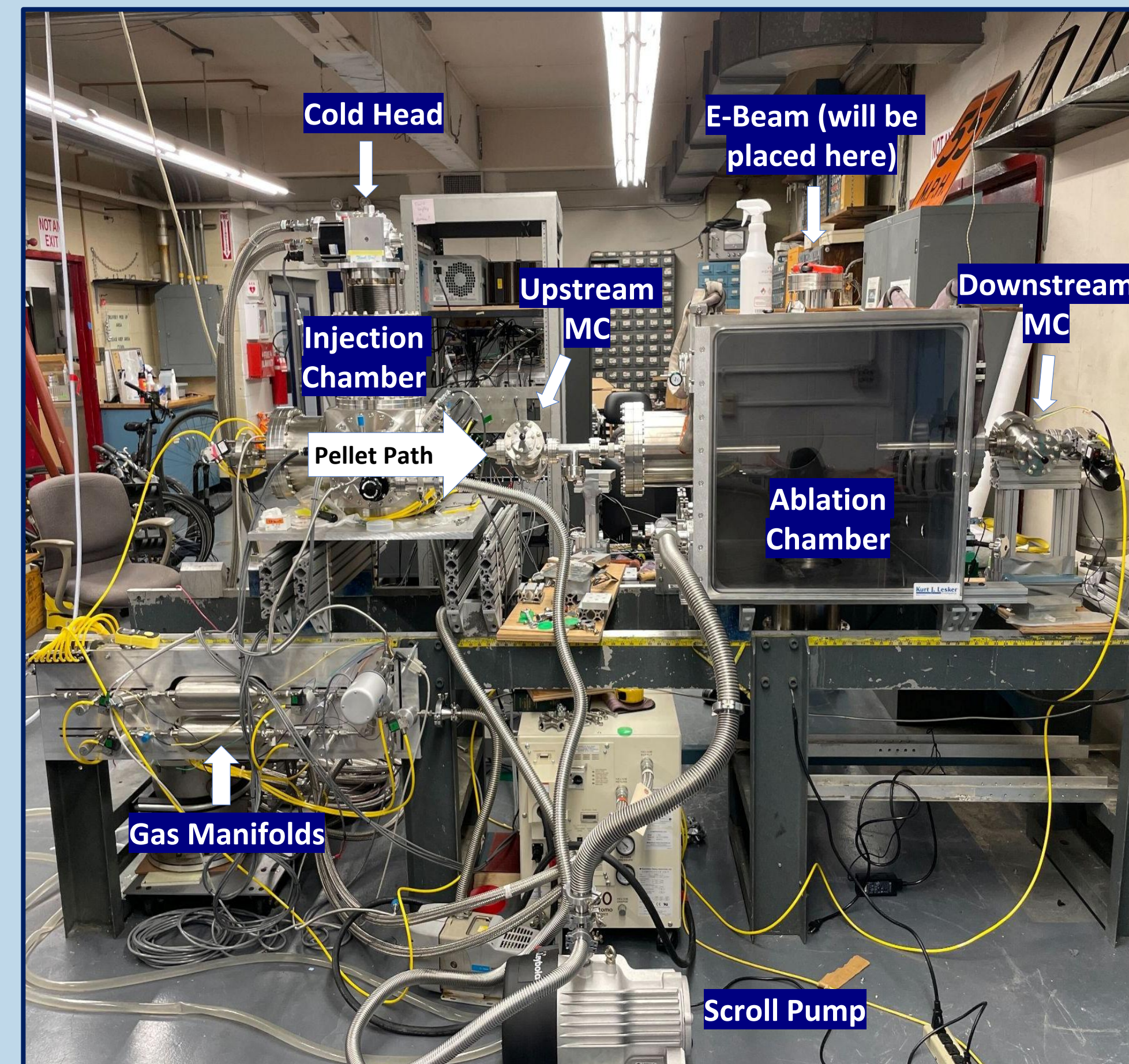
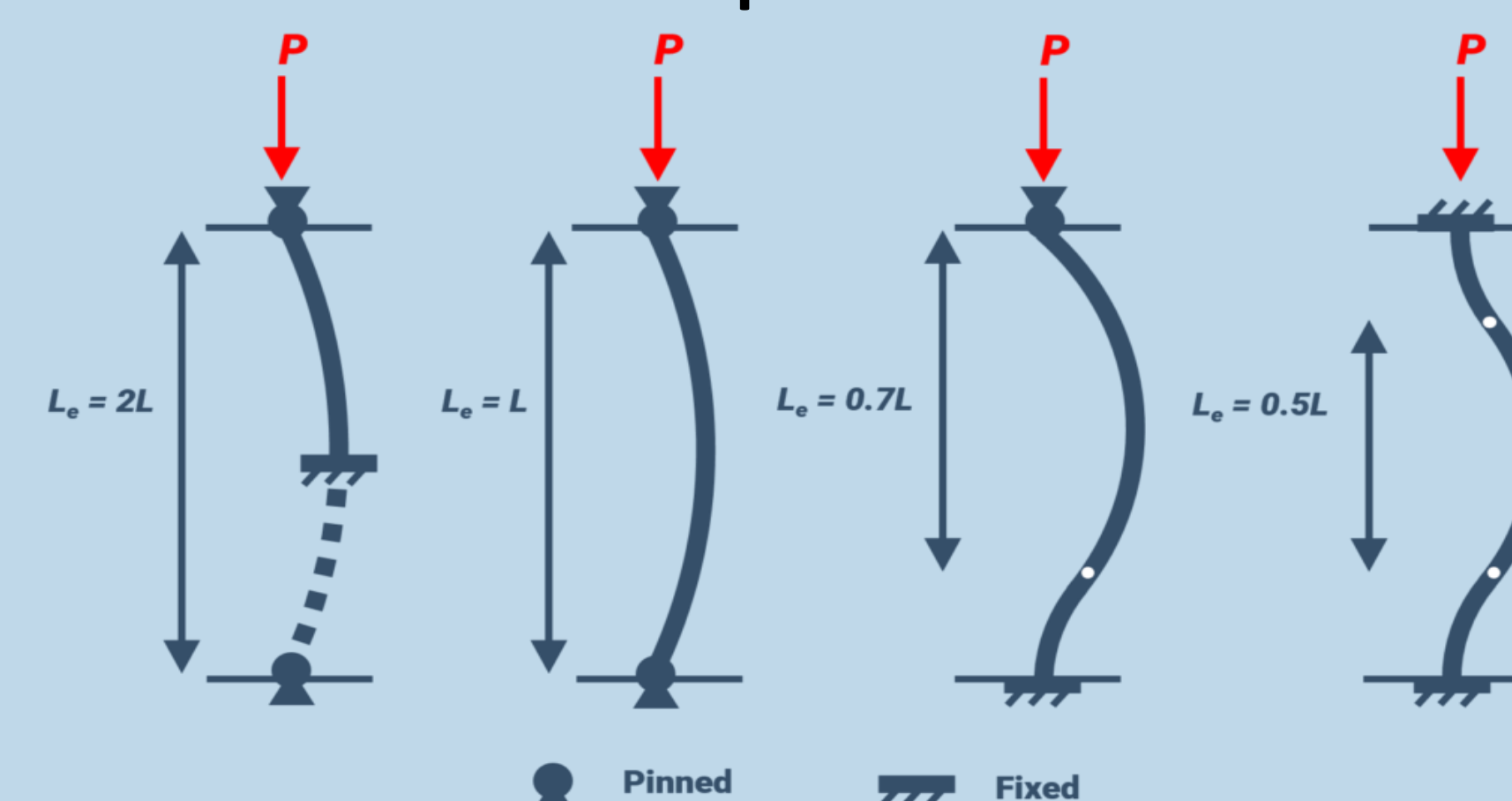


Fig 1. Labeled picture of the PAC hardware set-up including all relevant instruments and systems

Neon Pellets

- PAC's mechanical punch could buckle under the force needed to shear Neon pellets from the barrel



- Using Eulers Buckling force equation below, the buckling force of the punch is ~250lbs

$$F_B = \frac{\pi^2 EI}{(KL)^2}$$

- Where E is the modulus of elasticity, I is the second moment of inertia (Ar^2), K is the buckling constant, and L is the effective length
- This force is then compared to the shear force of the pellet, calculated through the following equation

$$F_{shear} = P_{shear} * 2\pi rL$$

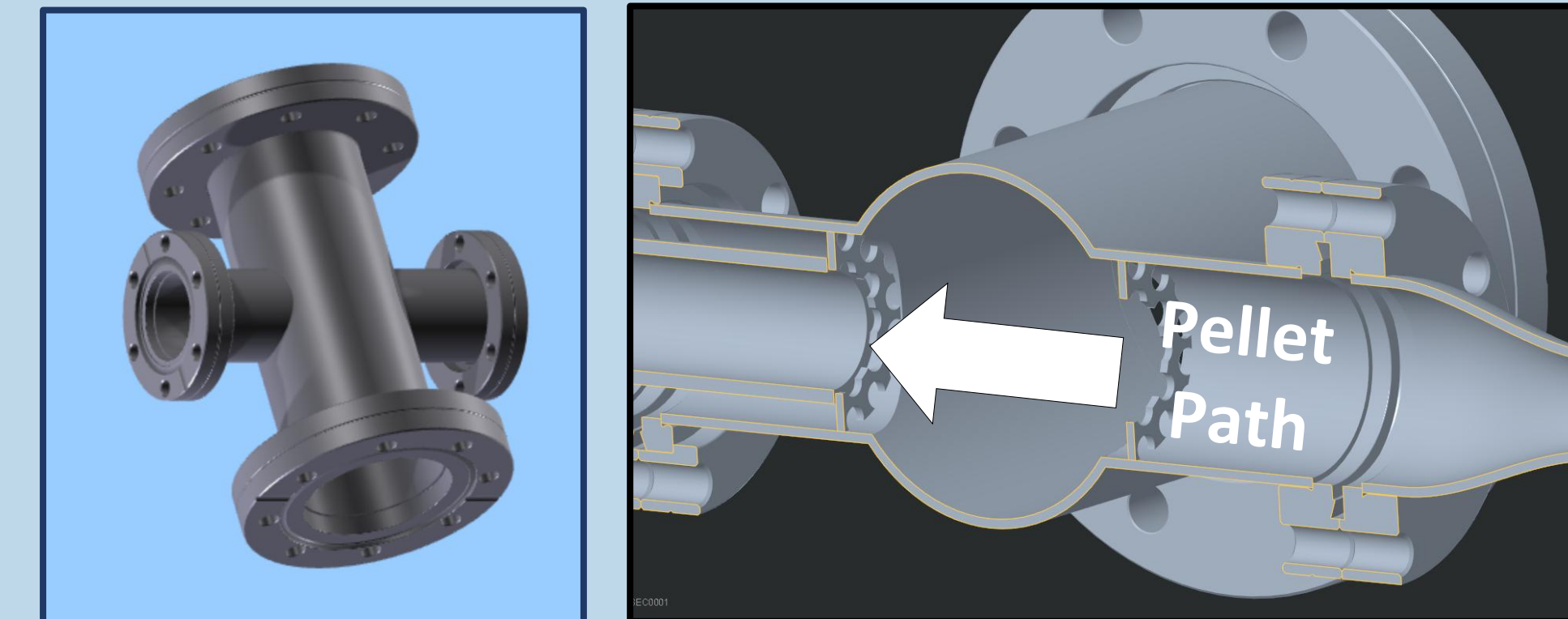
- Where the shear strength (P_{shear}) in MPa comes from the following relation [2]

$$P_{shear} = -0.1692 * T + 4.0463$$

- If shear force is greater than buckling force, the punch buckles

Microwave Cavities

Microwave Cavities (MCs) are used to find the mass of PAC's cryogenic pellets



HOW IT WORKS:

- The peak voltage of the oscillator is found within the MCs
- The oscillator is then tuned to ~60% of its peak voltage
- Pellets with a dielectric constant greater than the vacuum move through the cavities then MCs electric field changes
- Change in voltage on oscilloscope indicates a frequency shift in MCs, meaning a pellet has gone through

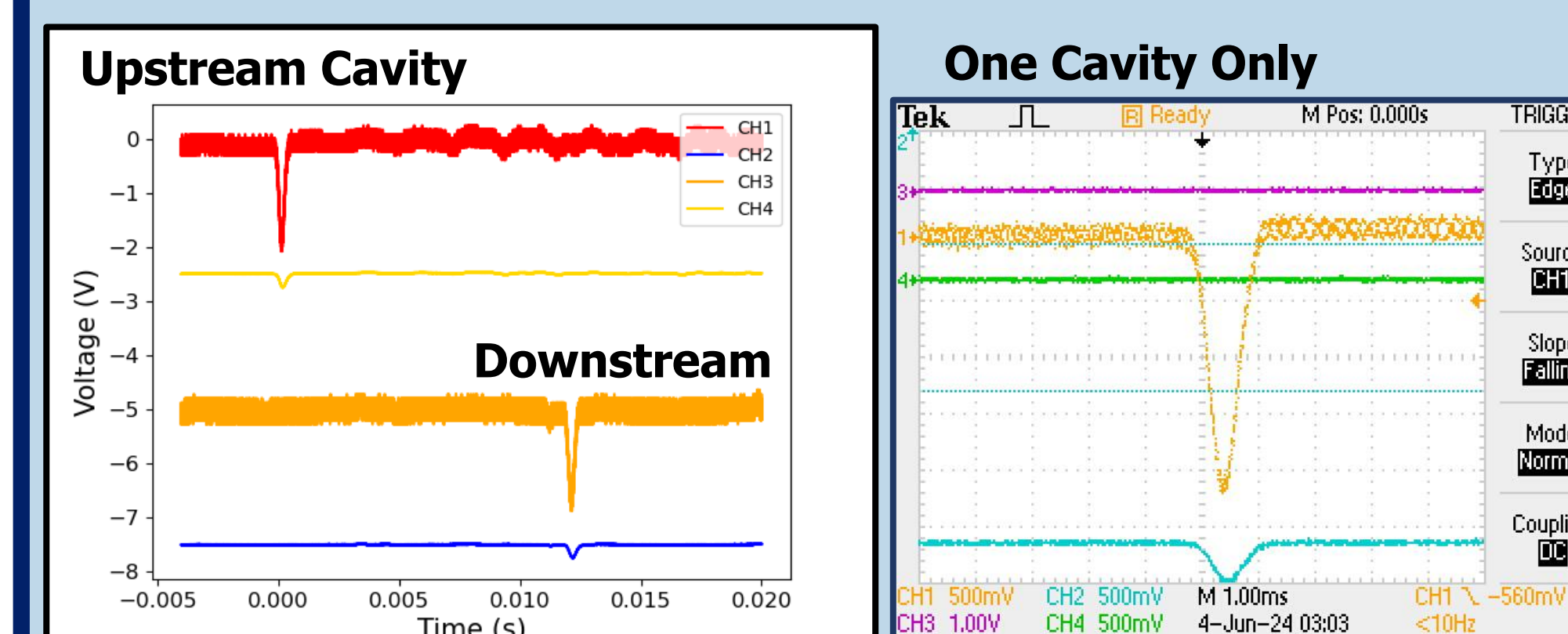


Fig 2. Plotted oscilloscope output of a pellet moving through the MCs. Left plot shows pellet moving through upstream and downstream MCs while right shows pellet moving through only one.

WHAT CALCULATIONS CAN BE DONE?

- Change in frequency within cavity is calculated by:

$$\Delta f_0 = \frac{\text{Pulse Out Voltage}}{\text{Cal Factor}} * \text{Detector Voltage}$$

- Pellet mass can be calculated by separating the pellet volume variable (below) and multiplying it by cryogenic density

$$\frac{\Delta f_0}{f_0} = \frac{-\left(\frac{3(e_r - 1)}{e_r + 2}\right) V_{Pellet}}{0.539 * V_{Cavity}}$$

- Where e_r is the cryogenic Hydrogen Dielectric Constant, V_{Pellet} is the volume of the pellet, V_{Cavity} is the volume of the MCs, and f_0 is the frequency inside the MCs
- Pellet masses through MCs and gas manifolds can be compared, shown in the next column

Microwave Cavity Results

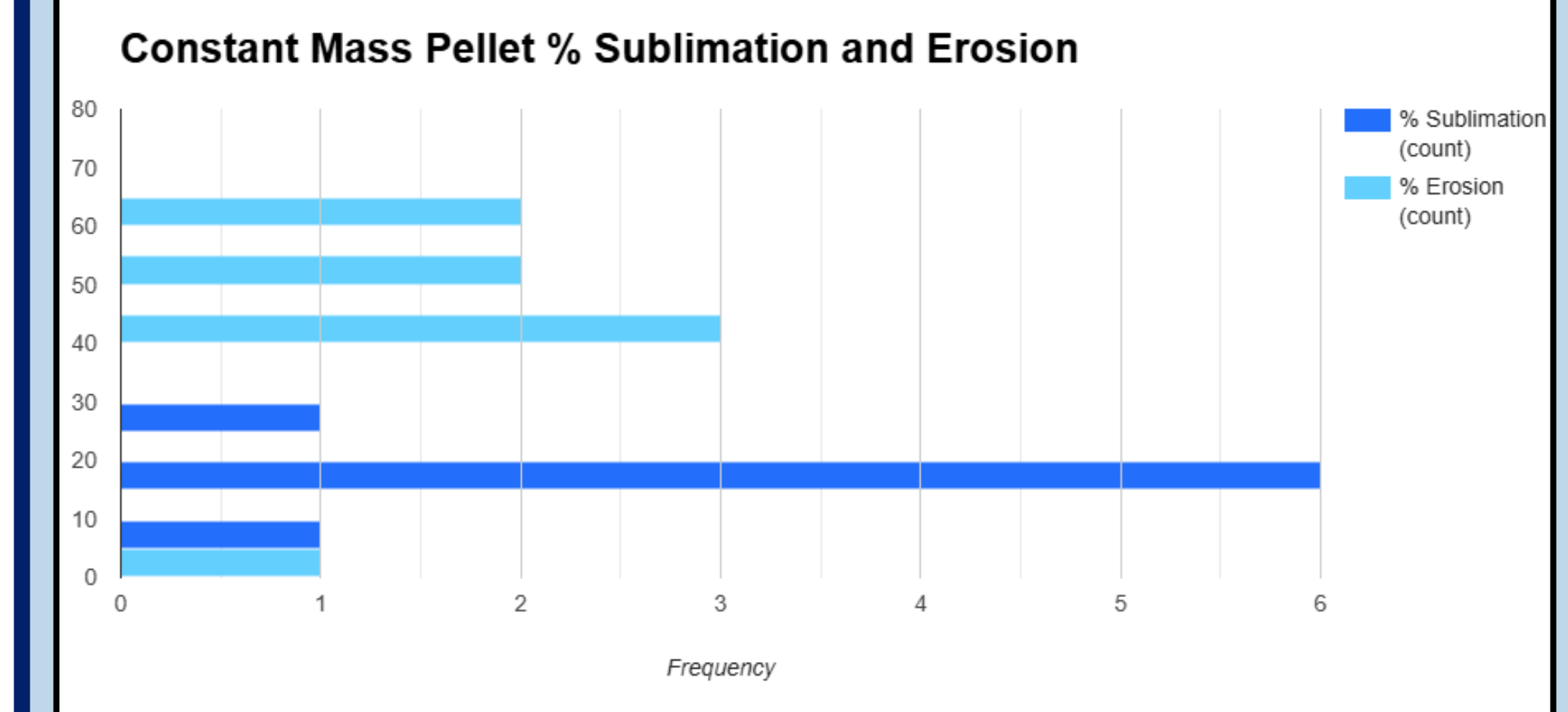


Fig 3. Histogram showing number of pellets for various % erosion and % sublimation values.

- 8 Pellets were shot at the constant starting amount of 250 Torr-L
- The percentage sublimated (difference between manifold calculated mass and MC mass 1) and percentage erosion (difference between MC mass 1 and 2) were calculated
- A standard deviation of $\pm 4.1\%$ for punch sublimation percentage and $\pm 18.9\%$ for erosion percentage were found
- High values of % Erosion indicate that pellet could be shattering before it reaches the downstream MC

Future Work

- Electron Beam will be installed in late August
- Punch will trigger the downstream gate valves to trap the pellet and recycle the gas after it sublimates
- Neon will be installed and tested once it arrives
- LabVIEW will be used for automation of the pellet-making process

References

- [1]: Combs, S. K., and L. R. Baylor. "Pellet-Injector Technology—Brief History and Key Developments in the Last 25 Years." *Fusion Science and Technology*, 73, no. 4, 2018, 493–518, DOI: 10.1080/15361055.2017.1421367.
[2]: Gebhart et al. (2021). Shear Strength and Release of Large Cryogenic Pellets from the Barrel of a Shattered Pellet Injector for Disruption Mitigation. *Fusion Science and Technology*, 77(7–8), 721–727. <https://doi.org/10.1080/15361055.2021.1874766>



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